1) Explain the difference between data and information

Data is raw, unprocessed facts like numbers, texts or measurements and information is processed, organized data that has a meaning. You can think of it like normal sentences. For example, "the average temperature is 25°C". So essentially data is what makes up useful and meaningful information.

2) What is metadata?

Metadata is data about data. It describes the structure, context, or properties of data so it can be understood and used properly.

For example, metadata might be file size or the date when a file was created, it might be column types in a database or even maybe the author of a document. Without metadata, data becomes difficult to interpret and pull apart.

3) What is a DBMS, and what are its advantages?

A DBMS (Database Management System) is software used to create, manage, and interact with databases. Examples include MySQL, PostgreSQL, SQLite.

The advantages of a DBMS is that, first of all, you've got data organization because you have a structured place to store the database. It reduces duplication and errors if you interface with it and if it's built correctly. It controls access to data, like maybe you have an API in place that needs to work with this data and the DBMS will control the access to that. It also means multiple users can access this data safely and if you have a database in something like cloud computing that means it's quite well protected against data loss.

4) Compare and contrast operational databases with analytical databases, and provide an example of each

Some differences between operational and analytical databases are that operational databases are for day-to-day operations. It stores current and detailed data. The queries via CRUD are fast and simple and the users might be using them via apps and so forth. For analytical databases, the purpose is mostly for data analysis and insights. So if you're in data science, you might work with data in an analytical database. The data is often historical and aggregated. The queries are often complex and heavy and the users are often analysts and data scientists.

An example of an operational database will be something like MySQL and an example of an analytical database will be something like Amazon Redshift, which is a big data warehouse that analyzes sales trends.

5) Explain the types of data and use cases where NoSQL databases are most effective

Different types of data are structured data with tables, semi-structured data which is JSON, and completely unstructured data like images and videos in your Google Drive. NoSQL databases like MongoDB and Cassandra work best when the data is unstructured or semi-structured or when you need flexibility, like the schema can change, or you need to handle large-scale or distributed data, or you need high read and write performance. The use cases for NoSQL are something like social media, or real-time analytics, or IoT data streams, or content management systems. So, in other words, if your data doesn't fit neatly into tables, you need to use NoSQL.

6) Which DBMS does not require server configuration, and what are the advantages of using it?

SQLite does not require server configuration. The advantages of this are that obviously there's no server setup that's required. It's lightweight, it's fast, it can store an entire database in a single file, it's easy to embed, and it's also fantastic for local applications or mobile apps and prototyping. So when you don't need a full production level database server, you can use SQLite.

7) Explain ACID properties in the context of a DBMS

Acid properties ensure reliable database transactions:

Atomicity – all operations succeed or none do
(“all or nothing”)
Consistency – data stays valid after a transaction
(follows rules/constraints)
Isolation – transactions don’t interfere with each other
(even if running at the same time)
Durability – once committed, data is permanent
(even after crashes)

In other words, these are properties that result in trustworthy transactions.